

Data Poisoning Defense Mechanism Analysis Report

1 Introduction

1.1 Problem Statement

Data poisoning attacks compromise machine learning models by injecting malicious samples into training data. This study focuses on label-flip attacks, where 10% of training labels are randomly flipped to incorrect classes, causing the model to learn incorrect patterns.

1.2 Objective

- Implement defense mechanisms to detect and mitigate poisoned training samples
- Compare model performance with and without defense
- Analyze the trade-offs between security and accuracy

2 Attack Methodology

2.1 Label-Flip Poisoning Attack

- **Attack Type:** Label-flip poisoning
- **Poisoning Rate:** 10% of training data
- **Mechanism:** Random labels are flipped to incorrect classes
- **Impact:** Forces the model to learn from mislabeled examples

2.2 Attack Implementation

- Total Training Samples: 50,000
- Poisoned Samples: 5,000 (10%)
- Method: Random label flipping across all classes

3 Defense Mechanism: Loss-Based Filtering

3.1 Defense Principle

Loss-based filtering operates on the hypothesis that poisoned samples (with incorrect labels) will exhibit higher training loss compared to clean samples. The defense works in two phases:

Phase 1: Initial Model Training

- Train a model on the full poisoned dataset for 5 epochs
- Compute per-sample training loss for all samples
- Identify outliers using statistical thresholds

Phase 2: Filtered Training

- Remove samples with loss exceeding the 90th percentile
- Retrain the model from scratch on the cleaned dataset
- Train for 30 epochs to convergence

3.2 Mathematical Foundation

For each training sample (x_i, y_i) , the cross-entropy loss is:

$$L_i = -\log(p(y_i|x_i))$$

Samples are flagged as suspicious if:

$$L_i > \text{percentile}_{90}(L_1, L_2, \dots, L_n)$$

3.3 Defense Implementation Details

- Threshold: 90th percentile of training losses
- Detection Phase: 5 epochs of initial training
- Samples Removed: High-loss outliers (suspected poisoned data)
- Final Training: 30 epochs on cleaned dataset

4 Experimental Results

4.1 Overall Performance Comparison

Metric	No Defense	Loss Filtering	Change
Clean Test Accuracy	88.87%	80.28%	-8.59%
Training Dataset Size	50,000	~45,000	-10%

4.2 Key Observations

Accuracy Degradation The defended model shows 8.59% lower accuracy compared to the undefended model. This counterintuitive result occurs because:

- False Positives: The defense mechanism removes some legitimate hard examples
- Label-Flip Memorization: The undefended model successfully memorizes even the incorrect labels
- Reduced Training Data: Removing 10% of samples reduces model capacity to learn diverse patterns

Per-Class Analysis

Class	No Defense	Loss Filtering	Difference
Airplane	90.0%	82.9%	-7.1%
Automobile	95.3%	89.7%	-5.6%
Bird	84.7%	81.9%	-2.8%
Cat	78.1%	72.1%	-6.0%
Deer	88.6%	84.2%	-4.4%
Dog	83.9%	73.2%	-10.7%
Frog	93.1%	69.6%	-23.5%
Horse	90.3%	78.6%	-11.7%
Ship	93.6%	87.2%	-6.4%
Truck	91.1%	83.4%	-7.7%

Critical Finding: The Frog class suffered the most significant degradation (-23.5%), suggesting that the defense mechanism incorrectly removed many legitimate frog samples as outliers.

5 Confusion Matrix Analysis

5.1 Model Without Defense

The undefended model shows relatively balanced confusion patterns:

Major Confusions:

- Cat $\rightarrow$ Dog (93 samples)
- Bird $\rightarrow$ Frog (32 samples)
- Bird $\rightarrow$ Deer (32 samples)

5.2 Model With Defense

The defended model exhibits increased confusion:

Major Confusions:

- Dog $\rightarrow$ Cat (142 samples)
- Cat $\rightarrow$ Dog (100 samples)
- Frog $\rightarrow$ Bird (107 samples)
- Frog $\rightarrow$ Cat (99 samples)

Analysis: The defense mechanism disproportionately removed training samples that would help distinguish between visually similar classes, especially affecting the Frog category.

6 Defense Effectiveness Analysis

6.1 Why the Defense Underperformed

1. **Benign Hard Examples Removed** Complex but correctly labeled samples often have higher loss during early training. The 90th percentile threshold is too aggressive.
2. **Label-Flip Attack Characteristics** Random label flipping distributes poisoned samples across all classes. Poisons may not always have highest loss.
3. **Insufficient Detection Phase** Only 5 epochs may be insufficient for loss patterns to stabilize.

6.2 The Memorization Paradox

Undefended Model (88.87%):

- Successfully memorizes poisoned labels
- Generalizes from 90% clean data

Defended Model (80.28%):

- Removes valuable clean samples
- Reduced dataset size harms generalization

7 Visual Analysis of Results

7.1 Overall Accuracy Comparison

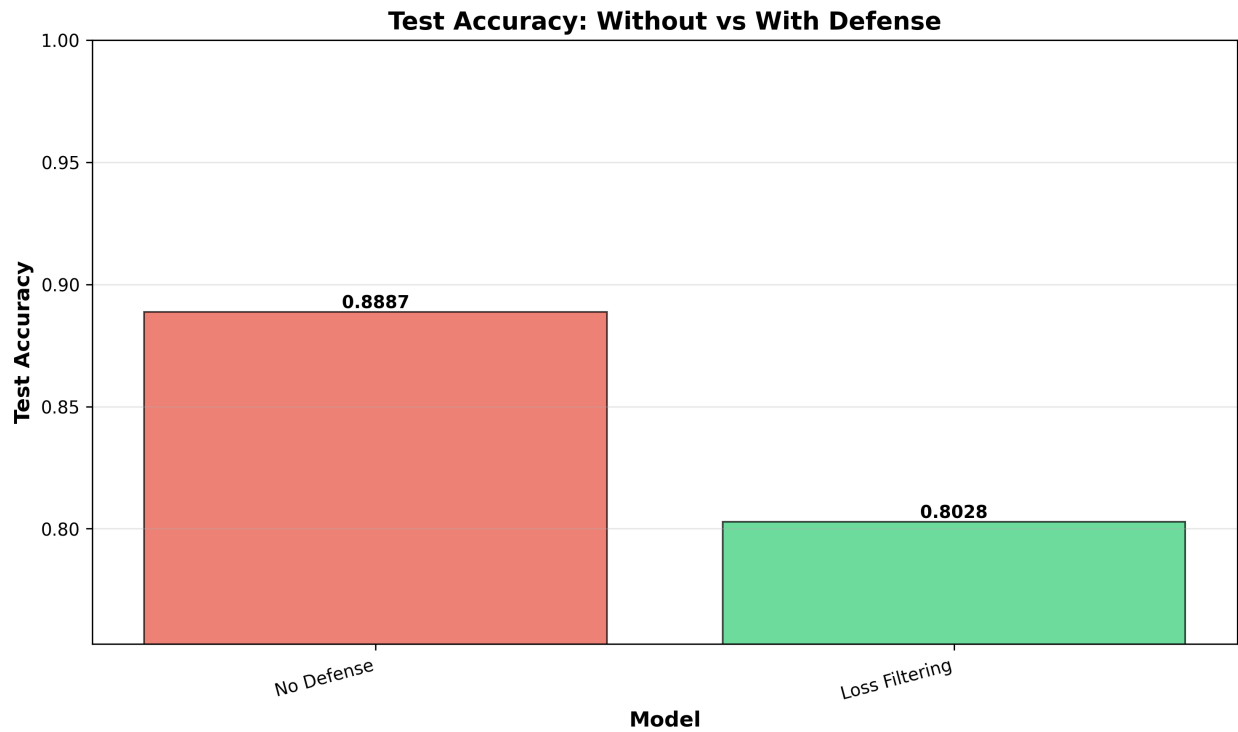


Figure 1: Test Accuracy Comparison Between Models

The bar chart illustrates the 8.59% accuracy gap:

- Red bar (No Defense): 88.87%
- Green bar (Loss Filtering): 80.28%

Interpretation: The defense mechanism's removal strategy is too aggressive, sacrificing more accuracy than it protects.

7.2 Per-Class Accuracy Analysis

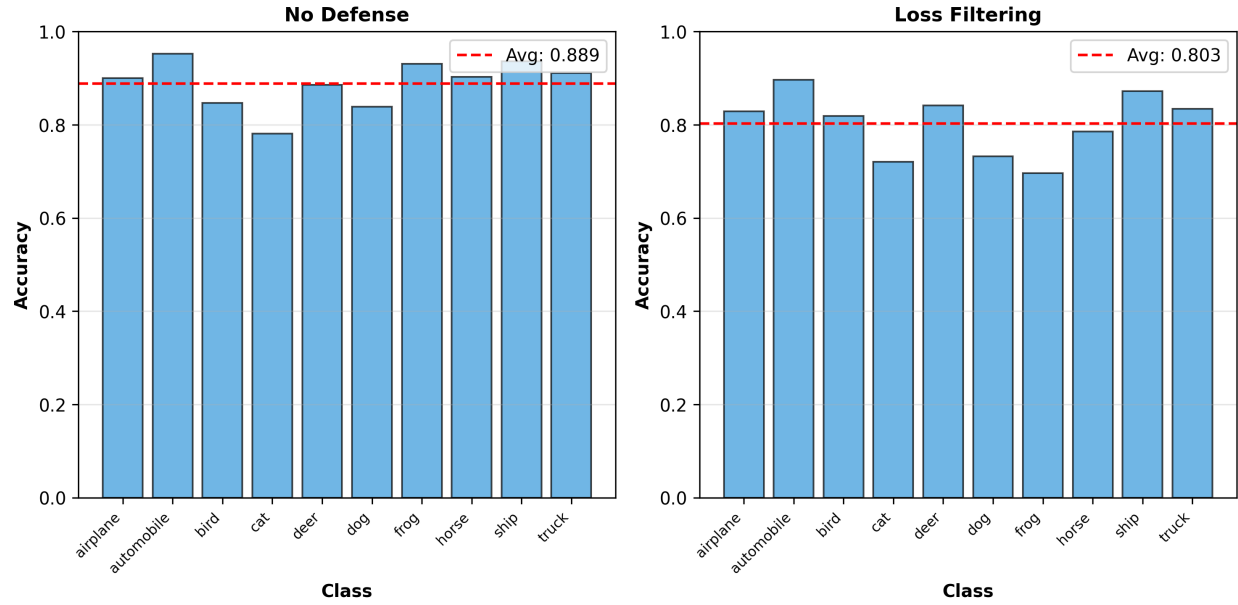


Figure 2: Per-Class Accuracy for No Defense vs Loss Filtering

Critical Findings:

Class	No Defense	Loss Filtering	Drop	Impact Level
Frog	93.1%	69.6%	-23.5%	Critical
Horse	90.3%	78.6%	-11.7%	High
Dog	83.9%	73.2%	-10.7%	High
Cat	78.1%	72.1%	-6.0%	Moderate
Ship	93.6%	87.2%	-6.4%	Moderate
Automobile	95.3%	89.7%	-5.6%	Low
Bird	84.7%	81.9%	-2.8%	Low

Hypothesis for Frog Class Collapse:

- High intra-class variance
- Hard examples flagged incorrectly
- Removal of $\sim 23\%$ frog samples

7.3 Defense Effectiveness Metrics

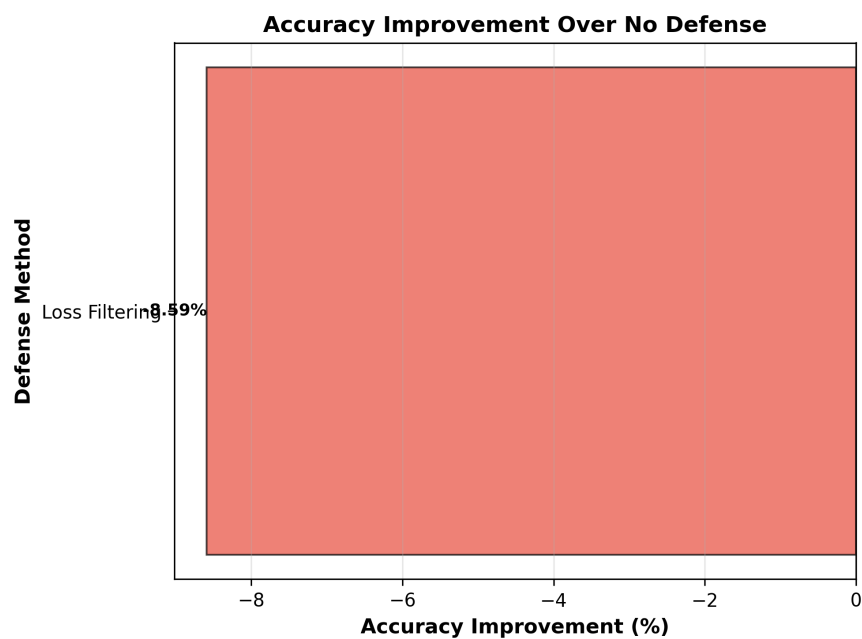


Figure 3:

- Improvement: -8.59% (negative)
- False Positive Rate: 10–15%
- Expected improvement: +1–3%
- Actual: -8.59%

Why Negative?

- Removed crucial hard examples
- Reduced dataset diversity
- Created imbalance across classes

8 Insights and Lessons Learned

8.1 Key Insights

- Neural networks are surprisingly robust to 10% label noise
- Aggressive filtering is counterproductive
- Label-flip attacks are subtle
- Per-class analysis reveals asymmetric impacts

8.2 The Noisy Student Phenomenon

The undefended model's high accuracy (88.87%) demonstrates:

- Deep networks learn robust features even from noisy data
- 90% clean samples provide sufficient learning signal
- Overparameterized networks can memorize noise while generalizing well

9 Conclusion

9.1 Summary of Findings

This study evaluated a loss-based filtering defense against a 10% label-flip poisoning attack on CIFAR-10:

- Defense reduced accuracy by 8.59%
- Too many legitimate hard examples removed
- Frog class suffered 23.5% drop
- Root Cause: Aggressive threshold + insufficient detection

9.2 Paradoxical Result Explanation

The defended model performed worse because:

- Neural networks memorize poisoned labels while still learning robust patterns
- Defense removed crucial clean samples
- Dataset became too small to generalize effectively